

Differential Equations

MAT 1032

Policy of Attendance:

Sufficient attendance is compulsory to sit the final examination

Please note that all tutorials should be submitted on the date(s) specified by the lecturer.

Method of Assessment:

End of semester theory examination	80%
Two (02) hour written examination	
Mid semester Examination	20%

References:

Elementary differential equations & Boundary Value Problems William E. Boyce,
Ricards C. Diprima.

Schaum's Solved Problems Series - Differential Equations

There are many other books that cover varies topics in differential equations. You should get into habit of using the library, where you should be able to find at least some books to use.

Differential Equations

independent variable x

dependent variable y

its derivatives

ordinary differential equations

Definition:

An equation involving derivatives of one or more independent variables is called a differential equation.

Definition. An equation involving **derivatives of one or more independent variables** is called a *differential equation*.

Examples

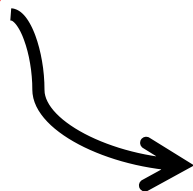
$$(1) \quad \frac{d^2 y}{dx^2} = -\omega y$$

$$(2) \quad \frac{d^2 y}{dx^2} - 2y = 0$$

$$(3) \quad \frac{dy}{dx} = \cos(2x + y)$$

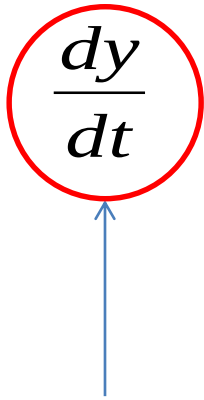
Differential Form

$$\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} = 0$$



Independent variable always on bottom

Derivative


$$\frac{dy}{dt}$$

Independent variable t and
dependent variable y ;

Rate of change

In our world things change and differential equations help us to understand the phenomena that involves rate of change.

Differential Equations...

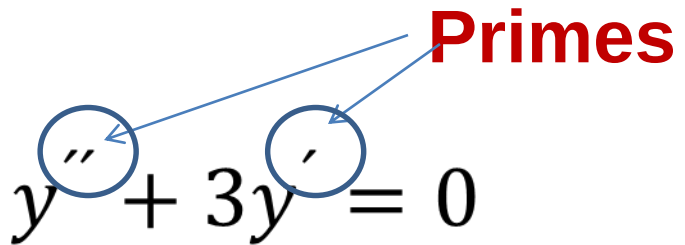
- **Derivative form**
- **Differential form**
- **Differential operator form**

Example:

$$y'' + 3y' = 0$$

Derivative Form

Primes



The diagram shows the equation $y'' + 3y' = 0$ with the second derivative symbol $''$ and the first derivative symbol $'$ each enclosed in a blue circle. Two blue arrows point from the word "Primes" to these circles.

$$y'' + 3y' = 0$$

Differential Form

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 0$$

Differential Operator

$$D^2y + 3Dy = 0$$

where $\frac{d}{dx} = D$

Solution of a differential equation

$$\frac{dy}{dx} = x$$

$$\Rightarrow \frac{dy}{1} = x \, dx$$

$$y = \frac{x^2}{2} + C \text{ is a solution.}$$

$$y = \frac{x^2}{2} \text{ is also a solution.}$$

**Families of
Solutions**

$$y = \frac{x^2}{2} + c$$

is the solution of the differential equation for every value of c , hence it is known as general solution.

Differential equations vs algebraic equations


$$\frac{dy}{dx} = \sin x$$

$$y = \cos x + c$$

function


$$x - 1 = 5$$

$$x = 6$$

number

Basic Concepts

Classification of differential equations

- Ordinary differential equations
- Partial Differential Equations
- Order
- Degree
- **Linearity** (Linear and non-linear differential equations)

Ordinary Differential Equation

Definition.

A differential equation involving derivatives with respect to a **single independent variable** is called an *ordinary differential equation*.

Examples

$$(1) \quad \frac{d^2 y}{dx^2} = -\omega y$$

$$(2) \quad \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = 0$$

$$(3) \quad dy = (x + \sin x) dx$$

Partial Differential Equations

Definition.

A differential equation involving partial derivatives with respect to **more than one independent variable** is called a *partial differential equation*.

Order of a Differential Equation

The **highest derivative** involved in a differential equation

$$\text{Ex) : (1) } \boxed{\frac{d^2 y}{dx^2}} - 5x \frac{dy}{dx} = e^x + 1 \quad \text{order : } 2$$

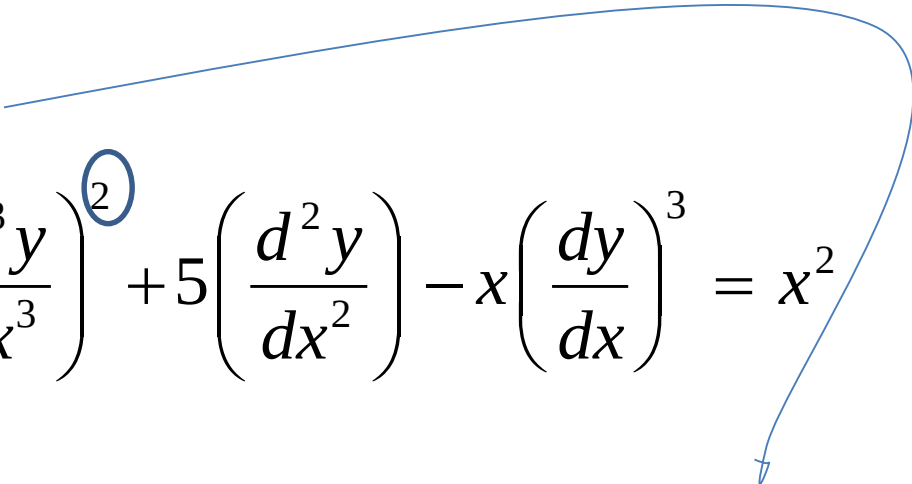
$$(2) \left(\boxed{\frac{d^3 y}{dx^3}} \right)^2 + 5 \left(\frac{d^2 y}{dx^2} \right) - x \left(\frac{dy}{dx} \right)^3 = x^2 \quad \text{order : } 3$$

Degree of a differential equation

Power of the highest derivative occurring in it after the equation has been rationalized with respect to derivatives.

$$\left(\frac{d^3 y}{dx^3}\right)^2 + 5\left(\frac{d^2 y}{dx^2}\right) - x\left(\frac{dy}{dx}\right)^3 = x^2$$

Order 3, degree 2



Formation of a Differential Equation

Example: (1)

Slope of any point (x, y) is proportional to x^2 .

Slope of any point (x, y) is $\frac{dy}{dx}$.

$$\frac{d x}{d t} \propto x^2$$

k proportionality constant.

$$\frac{d x}{d t} = k x^2$$

$$y = \int k x^2 dx$$

$$y = \frac{kx^3}{3} + c$$

Here c is an arbitrary constant.

Linear and non-linear Differential Equations

A differential equation is called *linear* if

- (i) every dependent variable and every derivative involved occurs in the **first degree only** and
- (ii) **no products of dependent variable and derivatives** occur. A differential equation which is not linear is called a *non-linear differential equation*.

(i) $\frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} = e^x + 1$ - **Linear differential equation.**

(ii) $x \left(\frac{dy}{dx} \right)^2 - y = e^x$ - **Non linear differential equation.**

(iii) $5 \frac{dy}{dx} - 3 \left(\frac{dy}{dx} \right)^7 y = x$ - **Non linear differential equation.**

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Geometrical interpretation of a differential equation

The method of direction fields is a graphical method for displaying the general shape and behavior of solutions to

$$\frac{dy}{dx} = f(x, y)$$

Direction Fields

Consider the first order D. E.

$$\frac{dy}{dx} = f(x, y)$$

- The equation specifies a slope at each point in the xy -plane where f is defined.
- It gives the direction that a solution to the equation must have at each point.

It does not require solving the differential equation. The uniform grid method and the isocline method are introduced, for computer and hand construction of direction fields.

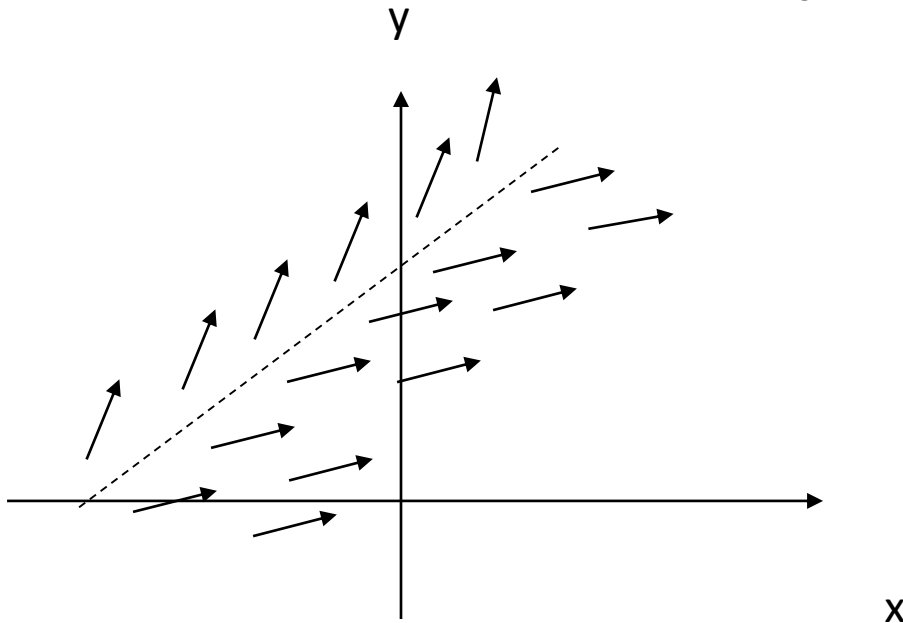
A plot of short line segments drawn at various points in the xy -plane showing the slope of the solution curve this is called a “**direction field**” for the differential equation.

The direction field gives us the “**flow of Solutions**”.

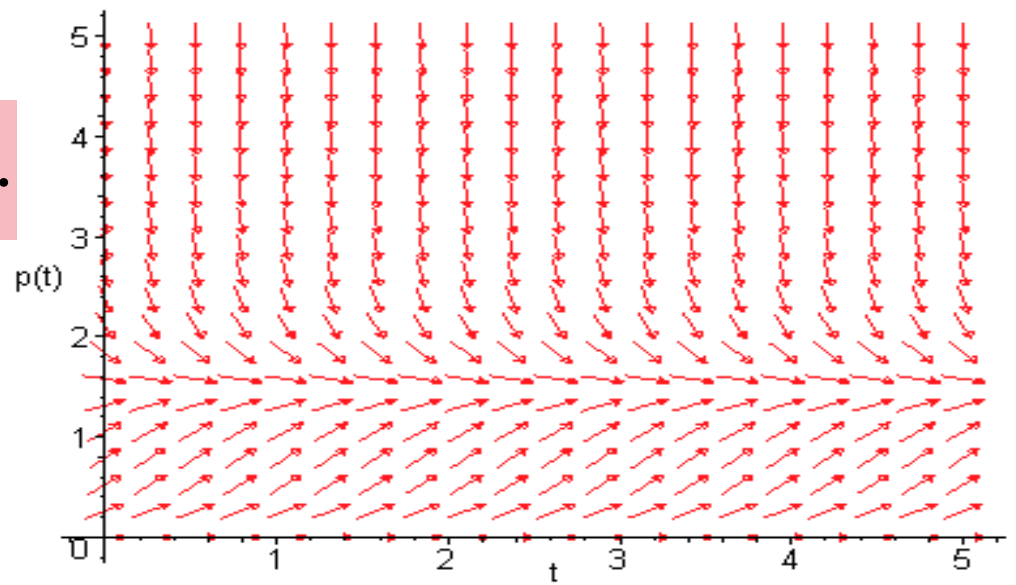
Differential equations and their stability

$$\frac{d y}{d x} = y - x$$

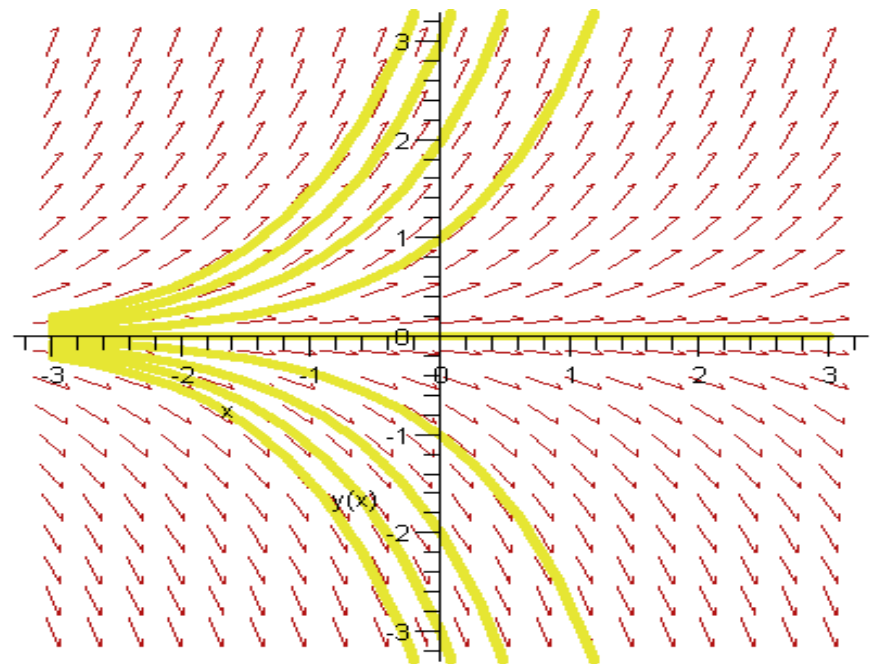
Direction field for the differential equation.



$$\frac{dp}{dt} = 3p - 2p^2, \text{ with } p(0) = 2.$$



$$y' = y$$



Exercises

$$\frac{d y}{d t} = (y - 1)$$

$$\frac{d y}{d t} = (1 - y)$$

$$\frac{d y}{d t} = y^2 - 5y + 6$$

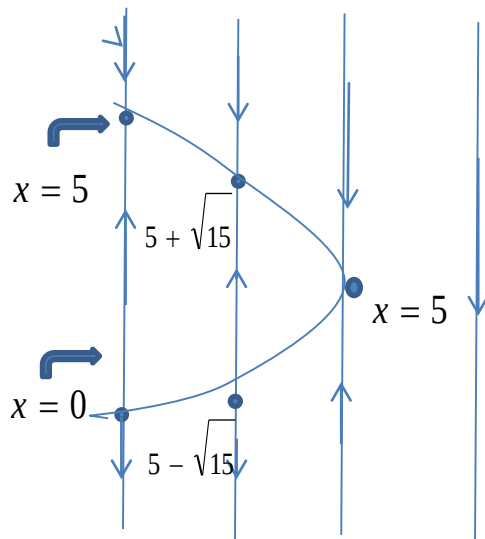
$$\frac{d y}{d t} = (y - 5)^2$$

$$\frac{d y}{d t} = 5y$$

Bifurcation –a way to study how an ODE depends on a parameter

$$\frac{d x}{d t} = \frac{1}{10} x(10 - x) - \underbrace{h}_{\text{Parameter}}$$

How depends on the parameter h



$$h = 0 \quad \frac{d x}{d t} = x(10 - x) = 0 \Rightarrow \boxed{x = 0, x = 10}$$

$$h = 1 \quad \frac{d x}{d t} = \frac{1}{10} x(10 - x) - 1 = 0 \Rightarrow \text{equilibria}$$

$$\boxed{x = 5 + \sqrt{15}, x = 5 - \sqrt{15}}$$

$$h = 2.5 \quad \frac{d x}{d t} = \frac{1}{10} x(10 - x) - 2 = 0 \Rightarrow \boxed{x = 5}$$

one equilibrium

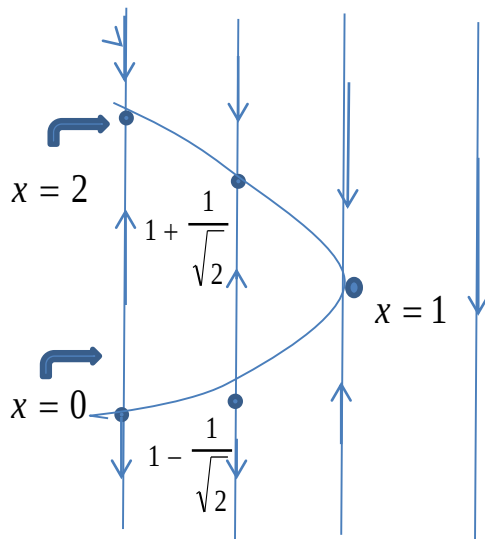
$$h = 3 \quad \text{no equilibria}$$

$$h = 0 \quad h = 1 \quad h = 2.5 \quad h = 3$$

Bifurcation –a way to study how an ODE depends on a parameter

$$\frac{d x}{d t} = x(2 - x) - \textcircled{h} \longleftarrow \text{Parameter}$$

How depends on the parameter h



$$h = 0 \quad \frac{d x}{d t} = x(2 - x) = 0 \longrightarrow \boxed{x = 0, x = 2}$$

$$h = \frac{1}{2} \quad \frac{d x}{d t} = x(2 - x) - \frac{1}{2} = 0 \longrightarrow \boxed{x = 1 + \frac{1}{\sqrt{2}}, x = 1 - \frac{1}{\sqrt{2}}}$$

equilibria

$$h = 1 \quad \frac{d x}{d t} = x(2 - x) - 1 = 0 \longrightarrow \boxed{x = 1}$$

one equilibrium

$$h = 3 \quad \text{no equilibria}$$

$h = 0 \quad h = 1 \quad h = 2 \quad h = 3$

Existence and uniqueness Theorem

Existence: Does a differential equation have a solution?

Uniqueness: Does a differential equation have more than one solution?